Shahaf Bassan

Ph.D. Candidate · Google PhD Fellow

Mechanistic Interpretability, Explainability, ML Verification, ML Theory, Robustness, Formal Methods, AI Safety

Hebrew University of Jerusalem

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Education

Hebrew University of Jerusalem — *Jerusalem, Israel*

Ph.D. in Computer Science — *January 2023–Current*

- Advisor: Prof. Guy Katz
- **Google PhD Fellowship** in Machine Learning and ML Foundations (2026–present)

Hebrew University of Jerusalem — *Jerusalem, Israel*

M.Sc. in Computer Science — *2021–2022*

- Advisor: Prof. Jeff Rosenschein co-advised with Dr. Yossi Adi

Hebrew University of Jerusalem — *Jerusalem, Israel*

B.Sc. in Computer Science and Physics — *2017–2020*

Publications

20 published or accepted papers, 17 of them as first, co-first, or last (supervising) author.
14 at top ML venues (5 ICLR, 4 ICML, 3 NeurIPS, 1 AISTATS, 1 TACL), **12 as first, co-first, or last author.**
6 at other peer-reviewed venues, incl. the top formal methods venues **CAV, TACAS, FMCAD**. ** equal contribution*

Idan Refaeli, Yizhak Elboher, Guy Katz, **Shahaf Bassan**. Do Feature Interactions Break Explainability? Provable Explanations for Any-Order Neural Additive Models *[**NeurIPS**, 2026]*

David Boetius*, **Shahaf Bassan***, Guy Katz, Stefan Leue, Tobias Sutter. Verified SHAP: Provable Bounds for Exact Shapley Values of Neural Networks *[**ICML**, 2026, Seoul]*[Paper]

Ron Eliav, Arie Cattan, Eran Hirsch, **Shahaf Bassan**, Elias Stengel-Eskin, Mohit Bansal, Ido Dagan. CLATTER: Comprehensive Entailment Reasoning for Hallucination Detection *[**TACL**, 2026, conditionally accepted]*[Paper]

Itamar Hadad, Guy Katz, **Shahaf Bassan**. Formal Mechanistic Interpretability: Automated Circuit Discovery with Provable Guarantees *[**ICLR**, 2026, Rio de Janeiro]*[Paper]

Shahaf Bassan*, Xuanxiang Huang*, Guy Katz. Unifying Formal Explanations: A Complexity-Theoretic Perspective *[**ICLR**, 2026, Rio de Janeiro]*[Paper]

Shahaf Bassan*, Yizhak Elboher*, Tobias Ladner*, Volkan Sahin, Jan Kretinsky, Matthias Althoff, Guy Katz. Provably Explaining Neural Additive Models *[**ICLR**, 2026, Rio de Janeiro]*[Paper]

Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, **Shahaf Bassan**, Guy Katz. FAME: Formal Abstract Minimal Explanation for Neural Networks *[**ICLR**, 2026, Rio de Janeiro]*[Paper]

Shahaf Bassan, Michal Moshkovitz, Guy Katz. Additive Models Explained: A Computational Complexity Approach *[**NeurIPS**, 2025, San Diego]*[Paper]

Reda Marzouk*, **Shahaf Bassan***, Guy Katz. SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism *[**NeurIPS**, 2025, San Diego]*[Paper]

Shahaf Bassan, Guy Amir, Meirav Zehavi, Guy Katz. What makes an Ensemble (Un) Interpretable? *[**ICML**, 2025, Vancouver]*[Paper]

Shahaf Bassan*, Yizhak Elboher*, Tobias Ladner*, Matthias Althoff, Guy Katz. Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations *[**ICML**, 2025, Vancouver]*[Paper]

Shahaf Bassan, Ron Eliav, Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons *[**ICLR**, 2025, Singapore]*[Paper]

Reda Marzouk*, **Shahaf Bassan***, Guy Katz, Colin de la Higuera.
On the Computational Tractability of the (Many) Shapley Values *[**AISTATS**, 2025, Mai Khao, Thailand]*[Paper]

Shahaf Bassan, Guy Amir, Guy Katz. Local vs. Global Interpretability: A Computational Complexity Perspective
*[**ICML**, 2024, Vienna, Austria]* ★ **Spotlight** *[Paper]*

Shahaf Bassan*, Shlomit Gur*, Sergey Zeltyn*, Konstantinos Mavrogiorgos*, Ron Eliav, Dimosthenis Kyriazis.
Self-Explaining Neural Networks for Business Process Monitoring *[**ICSBT**, 2025, Spain]* 🏆 **Best Paper Runner-up***[Paper]*

Guy Amir*, **Shahaf Bassan***, Guy Katz. Hard to Explain:
On the Computational Hardness of In-Distribution Model Interpretation *[**ECAI**, 2024, Santiago de Compostela]*[Paper]

Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan Refaeli, Guy Amir, Kyle Julian, **Shahaf Bassan**, Pei Huang, Ori Lahav, Min Wu, Min Zhang, Ekaterina Komendantskaya, Guy Katz, and Clark Barrett.
Marabou 2.0: A Versatile Formal Analyzer of Neural Networks *[**CAV**, 2024, Montreal, Canada]*[Paper]

Shahaf Bassan*, Guy Amir*, Davide Corsi, Idan Refaeli, Guy Katz.
Formally Explaining Neural Networks within Reactive Systems *[**FMCAD**, 2023, Iowa]* 🏆 **Best Paper Runner-up** *[Paper]*

Shahaf Bassan, Guy Katz. Towards Formal XAI: Formally Approximate Minimal Explanations of Neural Networks.
*[**TACAS**, 2023, Paris, France]*[Paper]

Shahaf Bassan, Yossi Adi, Jeffrey S. Rosenschein.
Unsupervised Symbolic Music Segmentation using Ensemble Temporal Prediction Errors *[**Interspeech**, 2022]*[Paper]

Fellowships and Scholarships

2026 **Google**, **PhD Fellowship in Machine Learning and ML Foundations**
2024 **Federman Security Center**, Federman Scholarship for Excellent PhD students
2023 **Federman Security Center**, Federman Scholarship for Excellent PhD students

Professional Experience

2023 **IBM Research**, Explainable AI Research Intern (Summer Internship)
Explainability during training; two papers, including *Explain Yourself, Briefly!* (**ICLR 2025**)
2019–2020 **Fiverr**, Software Engineer (key involvement in building “Fiverr for Business”)

Honors and Awards

2026 **Hebrew University of Jerusalem**, Chosen as 1 of 5 research students at HUJI university-wide, and the only CS & Engineering representative, to present my research to the Hebrew University Board of Governors at the “Journey to Science” research showcase
2026 **NeurIPS**, Outstanding reviewer award
2026 **ICML**, Outstanding (gold) reviewer award
2025 **ICML**, Outstanding reviewer award
2025 **ICSBT**, Best paper runner-up: *Self-Explaining Neural Networks for Business Process Monitoring*
2024 **FMCAD**, Best paper runner-up: *Formally Explaining Neural Networks within Reactive Systems*
2024 **ICML**, Spotlight paper award: *Local vs. Global Interpretability: A Computational Complexity Perspective*
2023 **Hebrew University of Jerusalem**, Outstanding teacher award
2023 **Hebrew University of Jerusalem**, Ranked first in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.
2022 **Hebrew University of Jerusalem**, Outstanding teacher award
2022 **Hebrew University of Jerusalem**, Ranked first in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.
2021 **Hebrew University of Jerusalem**, Outstanding teacher award

Invited Talks

October 2026 **Selected Speaker** at the Hebrew University's Board of Governors, "Journey to Science" research showcase (1 of 5 research students university-wide): "How Do You Know?"

December 2025 **Invited Keynote Speaker** at the Theory of Explainable ML Workshop, Copenhagen *[Workshop link]*

October 2025 **Invited Speaker** at the Theory of Interpretable AI Seminar: "The Computational Complexity of Explaining ML Models" *[Talk]*

June 2025 **Invited Speaker** at the IARCS Verification Seminar: "Formal XAI": Can we formally explain ML models? *[Slides]*

March 2025 **Invited Speaker** at the Interpretability Research Group, Tel Aviv: "Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons"

February 2025 **Invited Speaker** at Kings College, London: "Towards Formal Explainability of Neural Networks"

July 2024 **Invited Speaker** at Bosch Research Center for AI, Haifa: "Formal XAI: Formally Explaining the Decisions of ML Models"

September 2023 **Invited Speaker** at IBM Research, Haifa: "Formal XAI: Can we Formally Explain ML Models?"

June 2023 **Keynote Speaker** at Mobilit.AI, Toulouse, France: "Towards Formally Verifying and Explaining Deep Neural Networks" *[Talk]*

April 2023 **Invited Speaker** at TADM Workshop, Paris, France: "Formal Verification of Large Language Models"

April 2023 **Invited Speaker** at Live Workshop, Paris, France: "Formal Explainability of Deep Neural Networks"

Teaching Experience

2021–2023 **Data Structures and Algorithms**, Teaching Assistant.
2021, 2022, 2023: Chosen for the list of outstanding teachers at the Hebrew University.
2022, 2023: Ranked **first** in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.

Mentoring and Student Supervision

Current **Shimon Levy-Afgin**, M.Sc. student, continuing to a Ph.D.

Current **Lihen Laski**, M.Sc. student

Itamar Hadad, M.Sc. student, now a Ph.D. student (co-author of *Formal Mechanistic Interpretability*, ICLR 2026)

Joel Schreiber, M.Sc. student

Community Service

2025–Present **Theory of Interpretable AI seminar (an international monthly seminar on the theoretical foundations of interpretability)**, Co-Organizer

2025, 2026 **Selected as an outstanding reviewer**, ICML 2025, ICML 2026 (gold), NeurIPS 2026

2024–Present **ICLR, ICML, NeurIPS, TACAS**, Reviewer

Volunteer Experience

2020–2024 **Alumni Forum Co-Leader and Key Contributor**, Tech2Peace, an NGO bringing Israelis and Palestinians together through technology and entrepreneurship

2021–Present **Programming Teacher and Lecturer**, Tech2Peace